

Predicting and Preventing Preeclampsia

Clinical definition, the two early-screening routes, and what each one requires

BADIRA | بادرة

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ABOUT BADIRA

BADIRA is a self-adaptive framework for predicting the risk of pregnancy complications. It builds a standardised profile of each woman, measures the quality of the data available to it, and selects the model and prediction settings that the data can actually support — so the framework adapts to the population it is given instead of assuming one.

Every result is returned as three signals together, never as a bare number:

Risk — how high the risk of the complication is.

Reliability — how far this particular result should be trusted.

Time — by when it matters, and whether there is still time to act.

BADIRA is not only a prediction tool. It is built to work inside the window where a prediction can still change the outcome — from the first antenatal visit to 16 weeks — because that is when low-dose aspirin can still prevent preeclampsia rather than merely describe it. Predicting is not the goal; predicting early enough to act is.

Trust is treated as a requirement rather than a feature. BADIRA states its confidence, explains its reasoning, and never presents an estimate as a diagnosis.

Know early. Act in time.

What each part of this report covers

The report joins three pieces of research that were written separately. They are not placed side by side; they are connected, because the thing they have in common is the thing that matters.

Part	Section	What it covers
1	What preeclampsia is, and what counts as a diagnosis	The clinical definition, the diagnostic criteria, and the 20-week line that separates prediction from diagnosis.
2	Route A — the risk-factor checklist	The history-only SMFM/ACOG checklist, the factors it uses, and the aspirin decision it leads to.
3	Route B — FMF first-trimester combined screening	The reference model: what it measures, what equipment it requires, and how the alternatives compare with it.
4	The two routes side by side	Same decision, different evidence — accuracy, requirements and availability compared.
5	Biomarkers	What research measures in order to predict preeclampsia before anything is clinically visible.

Figure 1 — The whole picture on one page

How preeclampsia is found — and where Badira sits

Two ways to find a woman at risk before she is sick. One way to confirm she is sick. Each needs different things to be available.

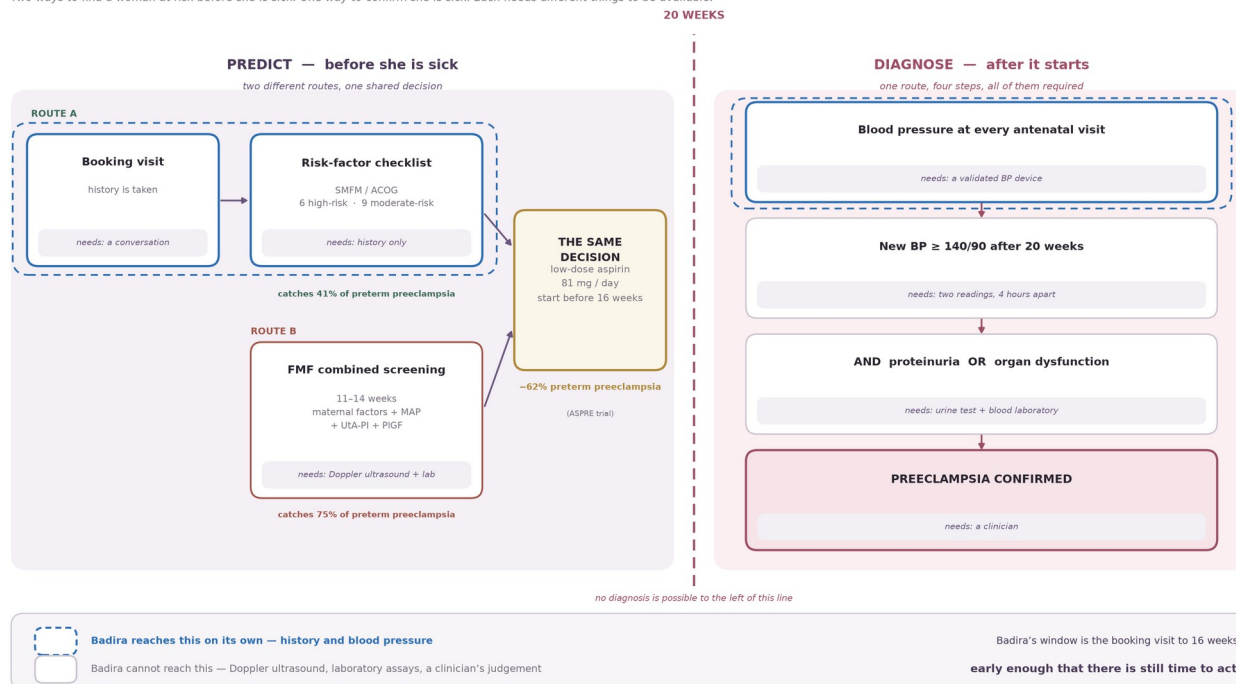


Figure 1. How preeclampsia is found. Two prediction routes converge on one decision before 20 weeks; one diagnostic chain runs after it. The dashed outline marks what Badira can collect without clinic equipment.

How to read this figure

Read it left to right, in four steps.

1. The red dashed line is 20 weeks. Left of it is prediction. Right of it is diagnosis. Nothing on the left is a diagnosis, because no guideline allows one before 20 weeks [1][2][3].
2. On the left there are two routes to the same place. Route A uses history only. Route B adds measurements. They need very different equipment — and they end in the same gold box: aspirin before 16 weeks.
3. On the right is the diagnosis, shown as a chain. Every step is required, and the last two need a laboratory.
4. The blue dashed outline is Badira. It marks the only three boxes Badira can fill on its own: the history, the checklist built from that history, and the blood pressure recorded at each visit.

The small grey line inside each box says what that step needs. That is the whole point of the picture: the two prediction routes reach the same decision, but they do not need the same things to get there.

Part 1 — What preeclampsia is, and what counts as a diagnosis

Before anything can be predicted, the thing being predicted has to be defined precisely. This part sets out that definition. It follows ACOG Practice Bulletin 222 [1] as its primary framework; NICE NG133 [2] and the ISSHP 2021 recommendations [3] differ in some details, so the guideline used should always be specified when defining labels for clinical data.

1.1 Preeclampsia in simple words

Preeclampsia is a pregnancy-related multisystem hypertensive disorder that usually develops after 20 weeks of pregnancy. It is more than “high blood pressure in pregnancy”: hypertension is accompanied by proteinuria and/or evidence of maternal organ involvement [1]. It can affect multiple maternal organs and systems, including the kidneys, liver, blood, brain, lungs, and blood vessels, and can also affect the placenta and baby. Placental dysfunction and maternal vascular dysfunction are important parts of the disease, although preeclampsia is complex and multifactorial. A woman can have clinically important preeclampsia without obvious symptoms, which is why routine antenatal monitoring is important.

Key concept: High blood pressure is an important sign of preeclampsia, but it is not the whole disease.

1.2 When can preeclampsia occur?

Preeclampsia typically develops after 20 weeks of pregnancy, and it is more common later in pregnancy. It can also develop after delivery, known as postpartum preeclampsia. Preeclampsia occurring before 20 weeks is very rare, rather than impossible, and should prompt investigation for unusual associated conditions [1].

Risk assessment can begin earlier in pregnancy, but it is important to distinguish risk of developing preeclampsia from an actual clinical diagnosis of preeclampsia.

1.3 Which organs and systems can be affected?

Because preeclampsia is a multisystem disorder, its effects are not limited to blood pressure.

System	What can happen
Blood vessels	Dysfunction of the blood-vessel lining can contribute to increased vascular resistance and fluid leakage.
Kidneys	Kidney involvement can cause protein to leak into the urine and kidney function to decline.
Liver	Liver injury can cause increased liver enzymes and may be associated with upper-abdominal pain.
Blood	Platelet levels can fall, and severe disease such as HELLP syndrome can involve breakdown of red blood cells.
Brain	Neurological involvement can cause headache or visual disturbances and, in severe cases, seizures (eclampsia).

System	What can happen
Lungs	Fluid can accumulate in the lungs (pulmonary edema), causing difficulty breathing.
Placenta / baby	Placental dysfunction can affect fetal growth and contribute to other pregnancy complications.

Important: These are possible effects of the disease. The specific organ findings used as clinical diagnostic criteria are covered separately in section 1.5.3.

1.4 Why preeclampsia matters

Preeclampsia matters because it can progress quickly and unpredictably, sometimes even when the woman initially feels well.

As the condition progresses, it can cause serious maternal complications, including eclampsia (seizures), stroke, HELLP syndrome, kidney or liver injury, pulmonary edema, and placental abruption. It can also affect the pregnancy and baby through complications such as fetal growth restriction, medically indicated preterm delivery, and, in severe cases, fetal or maternal death.

Another important challenge is that management often involves timing. Definitive management ultimately requires delivery of the placenta. Other treatments, such as antihypertensive medication and magnesium sulfate, are used when appropriate to reduce complications and manage risk while clinicians determine the safest timing of delivery [1].

This can create a clinical balance between allowing more time for fetal maturity and avoiding increasing maternal or fetal risk if the condition worsens. For this reason, early recognition, regular monitoring, and timely clinical assessment are important.

Preeclampsia also has implications beyond the pregnancy, as a history of the condition is associated with increased long-term cardiovascular risk.

1.5 The clinical diagnosis

Preeclampsia is diagnosed using a combination of blood pressure measurements and evidence of proteinuria and/or maternal organ involvement. The basic clinical framework is: new-onset hypertension after 20 weeks + proteinuria OR qualifying maternal organ dysfunction [1]. Therefore, hypertension is essential to the standard diagnostic framework, but proteinuria is not always required.

1.5.1 Blood pressure criteria

New-onset hypertension means that blood pressure becomes abnormal after 20 weeks of pregnancy in a woman who was previously normotensive.

- Systolic blood pressure ≥ 140 mmHg — the upper number, representing arterial pressure when the heart contracts.
- Diastolic blood pressure ≥ 90 mmHg — the lower number, representing arterial pressure between heartbeats.
- Systolic ≥ 140 mmHg or diastolic ≥ 90 mmHg = hypertension.
- Systolic ≥ 160 mmHg or diastolic ≥ 110 mmHg = severe-range hypertension.

Either value crossing its threshold is sufficient; both do not need to be elevated. Hypertension identified before 20 weeks is more consistent with chronic or pre-existing hypertension, although the woman's history and clinical context must be considered.

Ordinarily, an abnormal blood pressure is confirmed on two measurements at least 4 hours apart [1]. This is important because blood pressure can temporarily change because of factors such as anxiety, pain, physical activity, bladder fullness, incorrect cuff size, or white-coat effects.

An important ACOG distinction: if a woman has gestational hypertension and develops severe-range blood pressure, she is diagnosed and managed as having preeclampsia with severe features, even without proteinuria or another severe feature [1].

Severe-range blood pressure is particularly important because very high blood pressure creates an immediate maternal risk, including increased risk of stroke. Unlike ordinary hypertension, it can be confirmed within a short interval so that treatment is not delayed.

Blood pressure is one of the most important routinely measurable signs used to identify and monitor hypertensive disorders during pregnancy. It can help detect abnormal changes even when a woman has no obvious symptoms.

Because high blood pressure in pregnancy has several different causes, and they are not the same illness:

- A woman can have had high blood pressure before she was pregnant (chronic hypertension).
- She can develop high blood pressure during pregnancy with nothing else wrong (gestational hypertension).
- Or she can develop high blood pressure plus evidence that her organs are being affected — that is preeclampsia.

Two women can have identical blood pressure readings and have completely different conditions with completely different risks.

1.5.2 Proteinuria criteria

Proteinuria means protein is present in the urine. In preeclampsia, it reflects kidney involvement, as damage to the kidney's filtering system allows protein to leak into the urine.

Method	Diagnostic threshold
24-hour urine collection	≥ 300 mg per 24 hours
Protein-to-creatinine ratio (PCR)	≥ 0.3
Urine dipstick	2+ when quantitative methods are unavailable

1.5.3 Preeclampsia without proteinuria

A major point in the modern definition is that proteinuria is not required in every case of preeclampsia [1].

A woman with new-onset hypertension after 20 weeks can meet criteria for preeclampsia without proteinuria when qualifying new maternal organ dysfunction is present. Qualifying findings include:

- Thrombocytopenia: platelet count $<100,000/\mu\text{L}$. Falling platelets indicate involvement of the blood/clotting system.
- Renal insufficiency: serum creatinine $>1.1\text{ mg/dL}$ or doubling of the woman's baseline in the absence of other renal disease. The "doubling of baseline" criterion is important because a clinically meaningful decline in kidney function may occur even while the absolute creatinine value remains below 1.1 mg/dL .
- Impaired liver function: liver transaminases $\geq 2\times$ the upper limit of normal and/or severe persistent right-upper-quadrant or epigastric pain that is not otherwise explained.
- Pulmonary edema: fluid accumulation in the lungs that can impair breathing and oxygenation.
- Neurological or visual features: new-onset headache that is unresponsive to medication and not explained by another diagnosis, or visual symptoms not explained by another diagnosis.

This is important because clinically significant preeclampsia can occur with little or no proteinuria.

1.5.4 The diagnostic logic

1. Is there new-onset hypertension after 20 weeks?
2. If yes, is proteinuria present?
3. If proteinuria is absent, is qualifying maternal organ dysfunction present?

If the required accompanying findings are present, the clinical framework for preeclampsia can be met.

Key principle: Preeclampsia is not defined by blood pressure alone, and it is not defined by proteinuria alone. The diagnosis depends on the combination of hypertension with the appropriate accompanying clinical findings.

1.6 Distinguishing preeclampsia from other hypertensive conditions

Condition	Key distinction
Chronic hypertension	Hypertension present before pregnancy or before 20 weeks
Gestational hypertension	New hypertension after 20 weeks without proteinuria or qualifying organ dysfunction
Preeclampsia	New hypertension after 20 weeks with proteinuria and/or qualifying organ dysfunction
Preeclampsia with severe features	Preeclampsia with severe-range BP or qualifying severe features

These disorders can share the same finding — high blood pressure — but they are not the same condition. Correct classification depends on when the hypertension began, the woman's previous history and baseline, and whether additional features develop.

Key concept: A single blood-pressure reading does not tell the whole story. Pregnancy timing, previous measurements, and new clinical changes help determine what that reading means.

1.6.1 Superimposed preeclampsia

Superimposed preeclampsia occurs when a woman with chronic hypertension develops preeclampsia during pregnancy. It can be harder to recognize because her blood pressure may already be elevated and, in some cases, proteinuria may already exist.

Clinicians therefore need to consider change from the woman's own baseline, such as worsening hypertension or new clinical abnormalities.

This is an important example of why previous measurements matter: the meaning of a current value can depend on what was present earlier in pregnancy.

1.7 Symptoms, warning signs, and what does NOT diagnose preeclampsia

Preeclampsia may produce symptoms, but symptoms are not a substitute for the clinical diagnostic criteria described in section 1.5. Some women have no obvious symptoms, while many symptoms associated with preeclampsia can also occur for other reasons during pregnancy.

1.7.1 Symptoms that may occur

- Persistent or severe headache, particularly when it does not improve with usual treatment.
- Visual disturbance, such as blurred vision, flashing lights, spots, or temporary changes in vision.
- Pain in the right upper abdomen or epigastric area.
- New or unusual nausea and vomiting later in pregnancy.
- Sudden or marked swelling, particularly of the face or hands.
- Breathlessness or difficulty breathing.
- Reduced urine output.

Important: These symptoms should be taken seriously when clinically concerning, but none of them alone proves that a woman has preeclampsia.

1.7.2 Why symptoms alone cannot diagnose or exclude preeclampsia

1. Symptoms cannot exclude the disease. Preeclampsia can be present without obvious symptoms. A woman feeling well therefore does not rule it out.
2. Symptoms cannot diagnose the disease. Many possible symptoms are non-specific and can occur during pregnancy for reasons unrelated to preeclampsia.
3. Clinical context matters. Diagnosis depends on the defined clinical framework in section 1.5. Even symptoms that appear within diagnostic criteria may require additional qualification and professional assessment.

1.7.3 What does NOT diagnose preeclampsia

The following findings may be clinically useful or may prompt further assessment, but none should be treated as a diagnosis of preeclampsia by itself.

Finding	Why it is not a diagnosis by itself	What it represents
One high blood-pressure reading	A single non-severe reading requires appropriate confirmation and interpretation rather than being treated as a diagnosis on its own.	A clinical measurement that may prompt repeat measurement or assessment
Headache	Headache is common and can have many causes. The relevant neurological criterion is more specific than simply reporting a headache.	A symptom requiring clinical interpretation
Swelling (oedema)	Swelling is common during pregnancy and is not, by itself, a diagnostic criterion for preeclampsia.	A non-specific finding
Nausea	Nausea can occur for many reasons during pregnancy and is not diagnostic on its own.	A non-specific symptom
A risk factor or family history	Risk factors change the probability of developing preeclampsia; they do not establish that the disease is currently present.	An input to risk assessment
An AI risk score	A risk score estimates probability; it does not satisfy the clinical diagnostic criteria for preeclampsia.	A prediction or risk estimate
First-trimester risk screening	Early screening estimates future risk and may guide prevention or surveillance; it is not a diagnosis of current preeclampsia.	Screening / risk estimation

Key principle: A symptom, risk factor, screening result, single measurement, or AI output can contribute useful information, but it must not be presented as equivalent to a clinical diagnosis.

Part 2 — Route A: the risk-factor checklist

This is the first of the two prediction routes. It is the one that runs everywhere, because it needs nothing that a clinic might not have.

2.1 What this screening route is

Before preeclampsia develops, clinicians can flag women who have characteristics linked to a higher risk of developing it. The SMFM/ACOG checklist route is based mainly on maternal medical, obstetric and family history plus basic maternal characteristics [4][5].

Key point: this is risk assessment, not diagnosis. It asks who is more likely to develop preeclampsia later.

2.2 High-risk factors

High-risk factor	Meaning
Previous preeclampsia	Preeclampsia in a previous pregnancy
Multifetal gestation	Current pregnancy with twins or more
Chronic hypertension	Pre-existing chronic high blood pressure
Type 1 or type 2 diabetes	Pregestational diabetes
Renal disease	Pre-existing kidney disease
Autoimmune disease	For example systemic lupus erythematosus or antiphospholipid syndrome

Decision rule: If a woman has at least one high-risk factor, ACOG/SMFM recommend low-dose aspirin to help prevent preeclampsia, when appropriate [5].

2.3 Moderate-risk factors

Moderate-risk factor	Meaning
Nulliparity	No previous birth
Obesity	Prepregnancy BMI ≥ 30 kg/m ²
Family history	Mother or sister had preeclampsia
Low socioeconomic status	A social-risk factor in the 2020 checklist
Maternal age ≥ 35 years	Age 35 years or older
Prior low-birthweight / SGA history	Relevant previous low-birthweight or small-for-gestational-age history
Previous adverse pregnancy outcome	Relevant adverse outcome in a previous pregnancy
Interpregnancy interval >10 years	More than 10 years since the previous pregnancy
Race factor	Included as a moderate-risk factor in the U.S.-based checklist, reflecting social and healthcare disparities rather than biological differences

Decision rule: with more than one moderate-risk factor, low-dose aspirin should be considered/recommended according to the applicable ACOG/SMFM guidance [5].

2.4 How the factors combine

The two decision rules above are not independent statements about single factors. They are a counting rule, and a woman normally arrives with several factors at once. The table below works the rule through the combinations that occur most often, and shows what the same woman would look like to a combined model of the kind described in Part 3.

Combination she arrives with	High	Mod.	What the checklist decides	What a combined model would add
Chronic hypertension only	1	0	Aspirin recommended — one high-risk factor is enough	How high her pressure actually is today, and for how long
Twin pregnancy + obesity	1	1	Aspirin recommended — the high-risk factor alone triggers it; the obesity changes nothing	Combines both into one number, separating her from a woman carrying twins at a normal BMI
Previous preeclampsia + chronic hypertension + type 2 diabetes	3	0	Aspirin recommended — the same output as one factor alone	Places her far above a woman with a single high-risk factor, which the checklist cannot express
First pregnancy + BMI ≥ 30 + age ≥ 35	0	3	Aspirin recommended — more than one moderate factor	Weights how far above 30 the BMI is and how far above 35 the age is, rather than counting boxes
First pregnancy + family history	0	2	Aspirin recommended — more than one moderate factor	Same, plus her measured blood pressure
First pregnancy only	0	1	The checklist alone does not trigger aspirin	Still returns a risk number, which may be high or low depending on her measurements
No listed factor at all	0	0	She is not flagged	Can still return an elevated risk from blood pressure or biomarkers alone — this is the group the checklist misses

READ THE TABLE CAREFULLY — IT SAYS TWO THINGS

First: combination raises probability. It never produces certainty. A woman with three high-risk factors may have an entirely normal pregnancy, and a woman with none may develop preeclampsia. That is exactly why the output of this route is a preventive treatment and not a diagnosis — the checklist is allowed to be wrong in both directions, and aspirin is safe enough that acting on an uncertain signal is reasonable.

Second: look down the fourth column. Rows 1, 2 and 3 produce the identical decision from one factor, from two, and from three. The checklist cannot tell the difference between a woman at moderately raised risk and a woman at very high risk, because a checklist counts and does not compute. That single limitation is the reason Route B exists, and it is the gap Badira is aimed at.

2.5 What happens after a woman is flagged?

ACOG/SMFM recommend low-dose aspirin 81 mg/day for women who meet the risk-factor criteria. It is started between 12 and 28 weeks of gestation and ideally before 16 weeks, and continued until delivery [5].

Important: being high-risk does not mean the woman has preeclampsia or will definitely develop it.

2.6 What does this route require?

The checklist uses information that can be collected early in prenatal care: previous pregnancy history, current pregnancy information, medical history, family history, age, and prepregnancy BMI.

No specialized preeclampsia biomarker test or uterine artery Doppler is required for this checklist route. Those measurements belong to the FMF screening pathway in Part 3.

2.7 Main limitation

The checklist identifies increased risk but cannot tell with certainty who will develop preeclampsia. Its result also depends on having a complete and accurate history. A woman without recognized checklist risk factors can still develop the disease.

2.8 The updated 2026 SMFM checklist

SMFM published an updated checklist in 2026 that replaces the 2020 statement [6]. The main updates are:

- IVF is added as a moderate-risk factor.
- Sociodemographic factors are updated, emphasizing that increased risk associated with race reflects social inequities and racism rather than biological differences.

2.9 Clinical path in one view

1. Initial prenatal visit
2. Collect medical, obstetric and family history + maternal characteristics
3. Check high-risk and moderate-risk factors
4. ≥ 1 high-risk factor $\rightarrow$ low-dose aspirin recommended when appropriate
5. >1 moderate-risk factor $\rightarrow$ low-dose aspirin considered/recommended according to the applicable guidance
6. Continue antenatal surveillance
7. Risk screening $\neq$ diagnosis of preeclampsia

BRIDGE — HOLD ON TO STEP 4

Route A ends at one instruction: low-dose aspirin, started before 16 weeks. Keep that sentence in mind while reading Part 3, because Part 3 ends at the same place by a completely different road.

Part 3 — Route B: FMF first-trimester combined screening

This is the second prediction route, and the one usually described as the gold standard. It measures more, catches more, and needs more.

3.1 What is the FMF screening model?

The Fetal Medicine Foundation (FMF) model estimates a woman's risk of developing preeclampsia, especially preterm preeclampsia, at about 11–14 weeks. It combines maternal characteristics and medical/obstetric history with measured biomarkers [7][8].

Key point: The result is an individual risk estimate — not a diagnosis of preeclampsia.

3.2 What does it measure?

The main high-performance FMF first-trimester combination is maternal factors + mean arterial pressure (MAP) + uterine artery pulsatility index (UtA-PI) + placental growth factor (PIGF) [9].

PAPP-A can also be included, especially when already available from first-trimester screening, but FMF evidence shows that it does not significantly improve screening when PIGF is already included.

3.2.1 Maternal factors

These establish the woman's baseline risk. Examples include age, weight and height, previous pregnancy/preeclampsia history, chronic hypertension, diabetes, method of conception, family history of preeclampsia, and relevant autoimmune disease.

Needed: patient history and/or medical record. No special lab or imaging equipment.

3.2.2 Mean arterial pressure (MAP)

MAP represents average arterial pressure during one cardiac cycle. In FMF screening, blood pressure is measured using a standardized protocol and incorporated into the risk model after appropriate adjustment.

Needed: validated automated blood-pressure monitor, correct cuff size, and standardized measurement technique.

3.2.3 Uterine artery Doppler (UtA-PI)

This assesses resistance to blood flow in the uterine arteries supplying the uterus and placenta. The FMF protocol measures the pulsatility index (PI) of the left and right uterine arteries and uses their mean.

Needed: ultrasound machine with color and pulsed-wave Doppler + a trained/certified operator following the FMF protocol.

3.2.4 Placental growth factor (PIGF)

PIGF is a protein involved in placental blood-vessel development. Its level in maternal blood provides information related to placental function and is used as a biochemical marker in the model.

Needed: maternal blood sample + laboratory assay/platform that measures PIGF.

3.2.5 PAPP-A

PAPP-A (pregnancy-associated plasma protein-A) is a placental protein measured in maternal blood and is often available from first-trimester aneuploidy screening.

Needed: maternal blood sample + compatible laboratory assay.

Important: PAPP-A can be included, but the core high-performance FMF combination is maternal factors + MAP + UtA-PI + PIGF because adding PAPP-A when PIGF is already included does not significantly improve screening performance.

3.3 How are the measurements combined?

The measurements are not interpreted as separate diagnoses. The model starts with prior risk from maternal factors and history and combines it with the measured biomarkers to calculate a patient-specific risk. Biomarker measurements are standardized as multiples of the median (MoM) with appropriate adjustments before entering the model [9].

This is the sentence that separates Route B from Route A. Route A counts boxes. Route B multiplies a prior risk by measured evidence and returns one number for one woman.

3.4 Why this matters for the reach of any app

The FMF pathway shows that advanced screening needs more than information entered into an app. Maternal history can be collected digitally, but UtA-PI requires Doppler ultrasound and PIGF/PAPP-A require blood testing and laboratory assays.

3.5 Requirements at a glance

Component	Purpose	What is needed
Maternal factors	Baseline risk	History / medical record
MAP	Maternal blood pressure	Validated BP device + correct cuff + standardized protocol
UtA-PI	Uterine artery blood-flow resistance	Doppler ultrasound + trained operator
PIGF	Placental biomarker	Blood sample + laboratory assay
PAPP-A	Additional placental marker	Blood sample + laboratory assay

3.6 Other models, and why FMF is the reference

FMF is not the only first-trimester model for preeclampsia. Two other kinds exist, and neither has displaced it:

- The Fetal Medicine Barcelona (FMB) model, which uses similar inputs with different statistics. Direct comparisons place it slightly below FMF.

- Machine-learning models built on the same four inputs. In the comparisons published so far, none has beaten the FMF competing-risks model.

The third option is the history-only checklist of Part 2. It is far simpler to run and far less accurate; Part 4 sets the two against each other.

FMF is treated as the reference for two reasons. It has been externally validated in many more settings than any alternative, and it is the model the ASPRE trial used to choose which women received aspirin [10] — so it is the only one with trial evidence that acting on its result actually prevents the disease.

3.7 FMF pathway in one view

- 11–14 week visit
- Collect maternal factors/history
- Measure MAP
- Measure UtA-PI by Doppler ultrasound
- Measure PIGF (and PAPP-A if used/available)
- Combine inputs in FMF competing-risks model
- Output: patient-specific risk estimate
- High-risk result → clinical prevention/follow-up; NOT a diagnosis

Part 4 — The two routes side by side

4.1 Same decision, different evidence

	Route A — risk-factor checklist	Route B — FMF combined screening
What it uses	Medical, obstetric and family history plus basic maternal characteristics	The same history, plus MAP, uterine artery Doppler and PIGF
How it decides	Counts factors against a fixed list and applies a rule	Multiplies a prior risk by measured evidence and returns one patient-specific number
When	Any time from the booking visit	A fixed window, 11–14 weeks
What it needs	A conversation and a record. Nothing else.	A validated BP device, a Doppler ultrasound machine, a certified operator, and a PIGF laboratory assay
How much it catches	About 41% of preterm preeclampsia at a 10% false-positive rate (NICE criteria). The 2013 ACOG criteria catch about 5% at a 0.2% false-positive rate [11]	About 75% of preterm preeclampsia, and about 90% of early preeclampsia, at a 10% false-positive rate [9][11]
What it outputs	A yes/no flag	A probability
Where it can run	Any antenatal clinic in any country	Clinics with first-trimester ultrasound and access to a PIGF assay
What it leads to	Low-dose aspirin, before 16 weeks	Low-dose aspirin, before 16 weeks

4.2 Why both routes exist

The last row of the table is identical for both columns. The rows above it are not. That is the entire reason two routes exist: they buy the same treatment decision at very different prices, and with very different accuracy.

Route B is better at the job. In the ASPRE trial, women identified by the FMF model and given low-dose aspirin had about 62% less preterm preeclampsia than those given placebo [10]. That is a large effect for a cheap and widely available drug, and it is the evidence that makes early identification worth doing at all.

Route A is worse at the job and still recommended everywhere, because it is the only one that can be performed with no equipment. A checklist that catches four in ten is better than a Doppler machine that is not in the building.

4.3 What the difference actually is

A checklist counts. A model computes.

Section 2.4 showed this directly. A woman with one high-risk factor and a woman with three high-risk factors receive the same output from the checklist, because the rule is a threshold on a count. The FMF model gives them different numbers, because it treats each factor as evidence to be weighed rather than a box to be ticked.

This is also why the checklist misses the last row of Table 2.4 — the woman with no listed factor at all. She is invisible to a counting rule. She is not invisible to a model that also reads her blood pressure.

4.4 The availability problem

Route B needs four things in the same building at the same time: a certified sonographer, a Doppler machine, a PIGF assay, and a woman who presents between 11 and 14 weeks. Where all four are present, Route B should be used, and this report does not argue otherwise.

Where they are not all present — which is most antenatal care, in most of the world — the woman does not receive a worse version of Route B. She receives Route A, or she receives nothing at all until her blood pressure is measured at some later visit. The gap between the two routes is not a gap in accuracy. It is a gap in availability, and it is the gap this project is about.

Part 5 — Biomarkers: how research predicts preeclampsia

Parts 2 and 3 described what clinics do. This part describes what the underlying research measures, and why. It answers the question: are there biomarkers that detect the risk of developing preeclampsia?

5.1 What a biomarker is here

A biomarker in this context is something measurable in the mother's blood, months before the disease appears, whose level differs between women who will go on to develop preeclampsia and women who will not.

This is possible because preeclampsia begins in the placenta long before it shows up as high blood pressure. The placenta of a woman who will develop preeclampsia releases a different balance of proteins into her circulation from early pregnancy onwards. Biomarkers read that imbalance. They are measuring the cause, which is why they can speak earlier than blood pressure can.

The short answer: yes. The strongest are PIGF and sFlt-1, and the balance between them. But every one of them requires a blood sample and a laboratory assay — none can be collected by asking a woman a question.

5.2 The biomarkers used or studied

Biomarker	What it is	What happens in preeclampsia	Status
PIGF	Placental growth factor — a protein that drives placental blood-vessel growth	Falls early, before symptoms. A consistent early decrease is seen in women who later develop preeclampsia	In routine clinical use. Part of the FMF triple test (Part 3)
sFlt-1	Soluble fms-like tyrosine kinase-1 — a protein that blocks the action of PIGF	Rises as the disease develops. Not predictive in the first trimester, but strongly so in the second and third	In clinical use after 20 weeks, not before
sFlt-1/PIGF ratio	The balance between the two, expressed as one number	Rises sharply in the weeks before the disease becomes clinically apparent	The most robust single predictor found in the literature — but for short-term prediction after 20 weeks [12]
PAPP-A	Pregnancy-associated plasma protein-A, a placental protein	Decreased levels in the first trimester are associated with early-onset preeclampsia	Available, but adds little once PIGF is included (Part 3.2.5)
PP-13	Placental protein 13	Abnormally low in the first trimester in women who later develop early-onset preeclampsia and fetal growth restriction	Research stage. Appears useful for early-onset disease only
Soluble endoglin	A protein released by the placenta that interferes with blood-vessel signalling	Consistently elevated in women who will develop preeclampsia	Research stage

5.3 The two different jobs biomarkers do

This distinction matters, and it is often blurred. The biomarkers are not all doing the same thing.

	First trimester (11–14 weeks)	Second and third trimester (20–34 weeks)
The question being asked	Will this woman develop preeclampsia later in this pregnancy?	Is this woman, who already has symptoms, about to develop it in the next few weeks?
Which markers work	PlGF (low). PAPP-A and PP-13 have some signal. sFlt-1 does not work here	The sFlt-1/PlGF ratio
What it is used for	Deciding whether to start aspirin before 16 weeks	Ruling preeclampsia out over the next week, so a woman can safely be sent home
Typical performance	As part of the FMF triple test: about 75% of preterm preeclampsia at 10% FPR [9]	A ratio ≤ 38 gives a negative predictive value of about 99% over the following week [12]

The second column is prediction in the sense this project uses the word. The third column is triage of a woman who is already unwell, and belongs to the right-hand side of Figure 1.

5.4 What none of them can do

No biomarker diagnoses preeclampsia. A low PlGF at 12 weeks does not mean a woman has preeclampsia — she cannot have it at 12 weeks, by the definition in Part 1. It means her placenta is developing in a way that is associated with the disease appearing later. The diagnosis still requires the chain on the right of Figure 1: blood pressure after 20 weeks, plus proteinuria or organ dysfunction.

And no biomarker is free. Every marker in Table 5.2 requires drawing blood and running an assay on a laboratory platform. That is a clinic capability, not an app capability.

5.5 What this means for Badira

Badira does not measure biomarkers, and this report does not claim it will.

Every biomarker above sits outside the dashed outline in Figure 1. Badira works with the two inputs that are on the near side of that line — history and blood pressure — and the reason Part 5 is in this report is not to propose adding biomarkers. It is to make the boundary explicit, so that no reader assumes a capability the project does not have.

Sources

Sources 1–8 are the guideline and organisation documents used in Parts 1–3 and were read directly. Sources 9–13 supply the performance figures quoted in Parts 3–5 and were obtained through published summaries rather than the full texts; each should be confirmed against the primary paper before it is quoted in a defence.

Clinical definition (Part 1)

1. American College of Obstetricians and Gynecologists. Gestational Hypertension and Preeclampsia. Practice Bulletin No. 222. Obstetrics & Gynecology. 2020.
2. National Institute for Health and Care Excellence. Hypertension in pregnancy: diagnosis and management. NICE Guideline NG133.
3. International Society for the Study of Hypertension in Pregnancy. Magee LA, Brown MA, Hall DR, et al. The 2021 ISSHP classification, diagnosis & management recommendations for international practice. *Pregnancy Hypertension*. 2022;27:148–169.

Risk-factor screening (Part 2)

4. SMFM Special Statement: Checklists for Preeclampsia Risk-Factor Screening to Guide Recommendations for Prophylactic Low-Dose Aspirin. 2020.
5. ACOG/SMFM Practice Advisory: Low-Dose Aspirin Use for the Prevention of Preeclampsia and Related Morbidity and Mortality. 2021.
6. SMFM Special Statement: Updated Checklists for Preeclampsia Risk-Factor Screening to Guide Recommendations for Prophylactic Low-Dose Aspirin. 2026.

FMF screening (Part 3)

7. Fetal Medicine Foundation — Preeclampsia Screening / Certification.
8. Fetal Medicine Foundation — Assessment of Risk for Preeclampsia.

Performance figures (Parts 3–5)

9. O’Gorman N, Wright D, Poon LC, et al. Accuracy of competing-risks model in screening for pre-eclampsia by maternal factors and biomarkers at 11–13 weeks’ gestation. *Ultrasound in Obstetrics & Gynecology*. 2017.
10. Rolnik DL, Wright D, Poon LC, et al. Aspirin versus placebo in pregnancies at high risk for preterm preeclampsia (ASPRE). *New England Journal of Medicine*. 2017;377:613–622.
11. Chaemsaihong P, Sahota DS, Poon LC. First trimester preeclampsia screening and prediction. *American Journal of Obstetrics and Gynecology*. 2022;226(2S):S1071–S1097.
12. Zeisler H, Llurba E, Chantraine F, et al. Predictive value of the sFlt-1:PlGF ratio in women with suspected preeclampsia. *New England Journal of Medicine*. 2016;374:13–22.
13. Externally validated prediction models for pre-eclampsia: systematic review and meta-analysis. *Ultrasound in Obstetrics & Gynecology*. (Source of the FMF vs Fetal Medicine Barcelona comparison and the machine-learning comparisons quoted in section 3.6.)

Prepared for Task 1, Senior Project I. The three research documents this report combines — the clinical definition, the risk-factor checklist and the FMF screening notes — are reproduced here in full; no content from them has been removed or rewritten.